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CS-499

April 5, 2026

Milestone 4

The artifact I have chosen is the inventory app I made using android studio for CS-360: mobile architecture and programming. This android application allows users to log in to the application using a username and password, which brings them to a screen to track inventory items. Users can add, delete, and change items as they see fit, with notifications sent to their phone when inventory is low. This application was finalized in October of 2025.

In this scenario, I have chosen this artifact as upgrading the database to a RESTful API using Node.js would improve this application, especially given the purpose of it. In a warehouse, it would be better if all workers in a warehouse were able to update a centralized database. To accomplish this, I chose to use MongoDB Atlas to host the database, with Node.js and Express as the middleware layer to handle database operations. Retrofit was then used in my application to communicate with the API and enabling synchronization between multiple devices. This component showcases my skills in backend programming and database design, as I had to correctly use HTTP requests and responses to communicate with the backend for CRUD operations for inventory in a database. The course outcome I planned to meet was outcome three and one, and I believe I had met it. Switching from a local database to a centralized one solves an industry-specific goal, while managing trade-offs of a more complex system, as well as enabling collaboration between workers through the use of a cloud-based database.

In creating this enhancement, I ran into an issue that helped me learn more about permissions needed for android applications. Firstly, my recycler stopped showing the items after updating to an API with node.js, although it looked like it should work since the phone itself could connect to the local host and showed the added JSON data. What I found out was that android was blocking my HTTP request as I did not realize it needed permission in the android manifest xml.